HI ECOMMERCE

PRODUCT RETURN MANAGEMENT SYSTEM

FINAL PROJECT DOCUMENTATION

Business Analysis Portfolio Project

Prepared for: HI ECOMMERCE (Fictional Company)
Project focus: Improving the online product return experience
Document scope: Excludes Testing and UAT

1. Executive Summary

HI ECOMMERCE is a fictional e-commerce company created for this Business Analyst portfolio project. The project proposes a Product Return Management System to make the return journey clearer, more consistent, and easier for customers and support teams. The proposed solution allows customers to submit return requests online, checks eligibility against business rules, creates a return reference number, and provides return and refund status updates.

This document summarises the business problem, objectives, scope, stakeholders, proposed solution, process design, requirements, practice data analysis, and Agile approach. Testing and User Acceptance Testing (UAT) are excluded and remain future activities.

2. Company and Project Overview

Item	Description
Company	HI ECOMMERCE (fictional e-commerce business created for this portfolio project)
Project	HI ECOMMERCE – Product Return Management System
Project type	IT Business Analysis / process improvement
Business area	Online shopping, product returns, customer support, and refunds
Primary users	Customers, customer-support agents, returns operations, and finance/refund staff
Proposed outcome	A structured digital process for submitting, validating, tracking, and managing product returns

3. Business Problem

Customers may find the return process confusing when instructions are unclear, eligibility rules are difficult to understand, or return and refund updates are not easy to track. Support teams may also spend time answering repeated questions and manually checking return information.

HI ECOMMERCE proposes a centralised return management process to improve visibility, reduce avoidable manual work, and provide customers with clearer information. These are problem statements for the fictional business case, not verified findings about a real retailer.

4. Project Objectives

- Provide customers with a simple online way to request a product return.
- Apply clear return eligibility rules before a request is accepted.
- Generate a unique return request reference for tracking.
- Make return and refund statuses visible to customers and support teams.
- Use return data to identify categories and reasons that may need business attention.
- Define requirements and workflows to guide a future software implementation.

5. Project Scope

In Scope

- Customer return request submission and return-reason selection.
- Return eligibility validation using configurable business rules.

- Return request ID generation and status tracking.
- Return and refund status visibility.
- Basic reporting of return rates, product categories, and return reasons.
- Business requirements, process design, Jira backlog planning, and portfolio documentation.

Out of Scope

- Building and deploying a production-ready application.
- Integrating with live payment gateways, courier partners, or inventory systems.
- Processing actual customer refunds or collecting real customer information.
- Completing software testing, UAT, or production sign-off in this project stage.

6. Stakeholders

Stakeholder	Interest / Responsibility
Customer	Submits a return request and checks request/refund status.
Customer Support Agent	Assists customers and reviews return request information.
Returns Operations Team	Reviews returned products and updates return progress.
Finance Team	Reviews and processes eligible refunds according to policy.
Business Owner / Management	Approves business rules, priorities, and expected outcomes.
IT / Development Team	Uses approved requirements to design and build the proposed system.
Business Analyst	Elicits requirements, documents processes, manages the backlog, and communicates with stakeholders.

7. Proposed Solution

The proposed HI ECOMMERCE Product Return Management System provides a single, structured journey for customers and internal teams.

1. Customer signs in and opens an order.
2. Customer selects the product to return and chooses a return reason.
3. System checks the request against the return policy.
4. If eligible, the system creates a return request ID and confirms submission.
5. Customer can view the return status and reference number.
6. The relevant operations team updates return progress.
7. Refund status is updated when the refund process reaches the relevant stage.

8. AS-IS and TO-BE Process Summary

AS-IS – Example Current Process

For this portfolio exercise, the AS-IS process is a realistic example of a less structured return journey: a customer contacts support or searches for return instructions, shares order details, waits for eligibility confirmation, receives return instructions, sends the product back, and follows up separately for status or refund updates. This is an assumed example and is not a verified process used by a named retailer.

TO-BE – Proposed HI ECOMMERCE Process

In the proposed process, the customer starts a return from their order history, selects a reason, and submits the request online. The system validates eligibility, creates a reference number, displays the next steps, and

provides status visibility. Internal teams update progress through a consistent workflow, including the refund stage.

AS-IS example	TO-BE proposal
Instructions may be spread across channels.	Return journey is available through one online flow.
Eligibility may require manual checking.	System checks eligibility using defined business rules.
Customer may contact support for updates.	Customer can view return status and reference number.
Return information may be handled inconsistently.	Request details and status are recorded in one process.

9. Business Rules

Rule ID	Business Rule
BR-01	Customer must be signed in to submit a return request.
BR-02	A return request must be linked to a valid order and product.
BR-03	The product must meet the return window and eligibility conditions defined by HI ECOMMERCE policy.
BR-04	A return reason must be selected before submission.
BR-05	The system must generate a unique return request ID for each accepted request.
BR-06	Return and refund statuses must be updated as the request progresses.

10. Data Analysis Summary

A practice dataset containing 5,000 order records was used to demonstrate return analysis. The dataset is for learning and portfolio purposes; it must not be represented as official HI ECOMMERCE, Amazon, or India-wide operational data.

Measure	Practice dataset result
Total orders analysed	5,000
Returned orders	2,500
Not returned orders	2,500
Overall return rate	50%

Category-level return rates in the practice dataset:

Category	Orders	Returned	Return rate
Clothing	920	919	99.89%
Electronics	918	614	66.88%
Home	847	338	39.91%
Sports	586	215	36.69%
Toys	536	157	29.29%
Beauty	609	163	26.77%
Books	584	94	16.10%

The practice data shows particularly high return rates for Clothing and Electronics. The unusually high Clothing result should be checked for data quality and validated at product or SKU level before business decisions are made.

The most frequently recorded reasons include Wrong item (310), Size/fit issue (298), Changed mind (279), Missing parts (276), and Duplicate order (276). These patterns demonstrate possible areas for further investigation; they do not prove the causes of returns.

11. Reporting and Dashboard Approach

The proposed reporting approach uses Excel and Power BI to help stakeholders monitor return performance. Suggested dashboard views include:

- Total orders, returned orders, and overall return rate.
- Return rate by product category.
- Return count by return reason.
- Filters for product category, shipping type, and payment method.
- Trend analysis over time when reliable date-based data is available.

The current portfolio materials demonstrate analysis and dashboard planning. An Excel dashboard pack is not the same as a published Power BI report or a native Power BI .pbix file.

12. Agile and Jira Approach

The project uses a Scrum-style Jira backlog to organise the proposed work into epics and user stories. This demonstrates how business requirements can be divided into manageable pieces of work.

Epic	Purpose
Return Request Management	Enable customers to create and submit return requests.
Return Eligibility Validation	Check whether a product meets the return policy.
Refund Management	Track refund processing and status.
Return Tracking and Notifications	Provide request progress and customer updates.
Reporting and Analytics	Provide visibility of return trends and performance.

Example User Story

As a customer, I want to submit a product return request online so that I can return an eligible product without contacting customer support.

Example Acceptance Criteria

- The customer is signed in.
- The customer can select a product from an eligible order.
- The customer can select a return reason.
- The system validates the return request.
- The system displays confirmation and a unique return request ID when submission is successful.

13. Expected Business Benefits

- A clearer and more convenient return experience for customers.
- More consistent application of return eligibility rules.
- Better visibility of return and refund progress.
- Potential reduction in repeated status enquiries through self-service.
- More structured information for returns and support teams.
- Improved management insight through return reporting and analysis.

These are expected benefits of the proposed solution. They have not been measured or confirmed through implementation.

14. Assumptions, Risks, and Dependencies

Assumptions

- HI ECOMMERCE is a fictional company created for this portfolio project.
- The return policy, eligibility period, and refund rules will be confirmed by business stakeholders.
- The practice dataset is illustrative and may not represent real customer behaviour.
- The proposed solution will require access to order and product information.

Risks and Dependencies

- Unclear or changing return policies may affect eligibility validation.
- Incomplete or inaccurate order data may lead to incorrect return decisions.
- A live solution may require integration with payment, courier, and order systems.
- Customer and payment information must be protected through suitable access controls and security measures.
- Business stakeholders must approve requirements and process rules before development.

15. Future Activities

The following activities are outside the scope of this document and should be completed before a real system could be approved for release:

- Confirm business rules and detailed requirements with stakeholders.
- Create or develop a working application or clickable prototype.
- Execute functional and integration testing.
- Conduct User Acceptance Testing (UAT) with representative users.
- Resolve defects and obtain formal business approval.
- Prepare implementation, training, and support plans.

Testing and UAT status: Not included in this document. These activities are planned for a later project phase and have not been completed.

16. Conclusion

The HI ECOMMERCE Product Return Management System project demonstrates a Business Analyst approach to understanding a business problem, defining a proposed solution, documenting requirements, mapping processes, organising work through Jira, and analysing practice data. The proposed system aims to make product returns more transparent and manageable for customers and internal teams. The next step is to validate the requirements with stakeholders, build a working solution or prototype, and then complete testing and UAT.